Bit Shift Operations

Dr. Md. Rukonuzzaman

Associate Professor
Department of Electrical and Electronic Engineering
American International University-Bangladesh (AIUB)

Left Shift (<<) Operation

- **Shifts all bits in a number to the left by a specified number of positions**
- **Fills the rightmost positions with zeros**
- **Effectively multiplies by 2^n where n is the shift amount**

Original:	0	0	0	0	0	1	0	1	(5)
	↓ Shift left by 2								
Result:	0	0	0	1	0	1	0	0	(20)

Original bits

New/filled bits

`Example: 5 << 2 (shift left by 2 positions)`

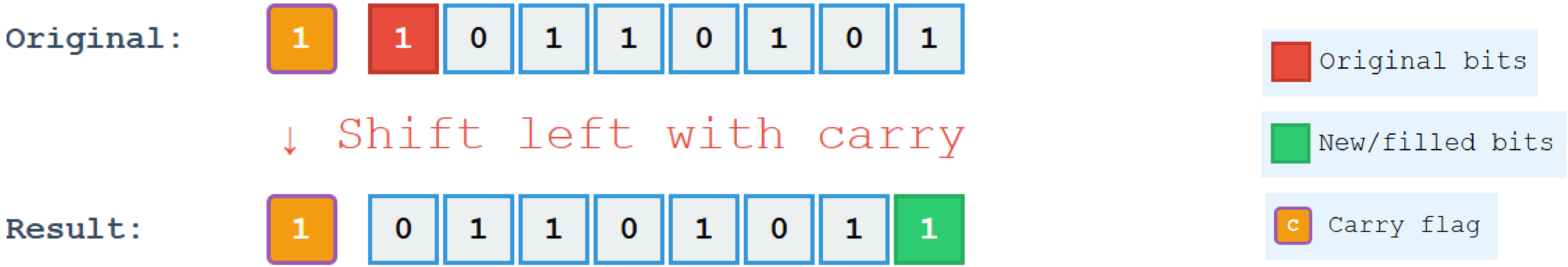
Right Shift (>>) Operation

- Shifts all bits in a number to the right by a specified number of positions
- Fills the leftmost positions with zeros
- Effectively divides by 2^n where n is the shift amount

`Example: 20 >> 2 (shift right by 2 positions)`

Left Shift (<<) with Carry

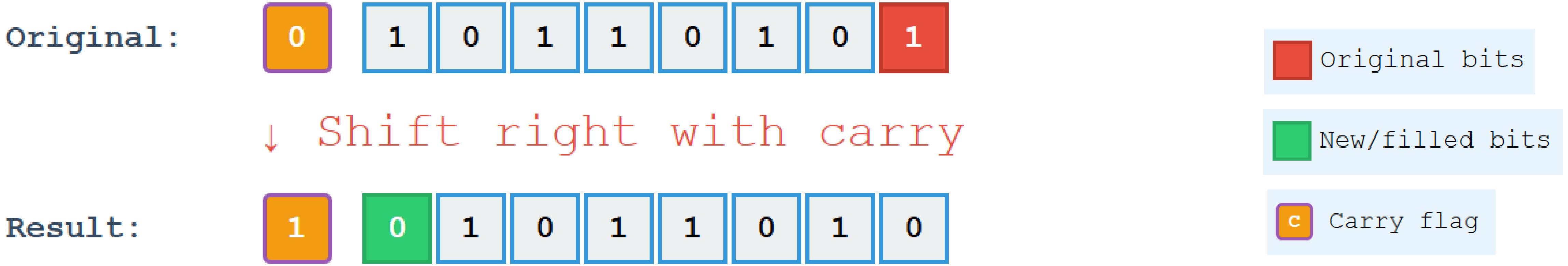
- Shifts all bits left by one position
- The leftmost bit (MSB) goes into the carry flag
- The previous carry flag value goes into the rightmost bit (LSB)
- Direction: MSB → Carry → LSB
- Useful for multi-precision arithmetic (working with numbers larger than register size), implementing rotate operations, serial data processing, checksum calculations



Example: Rotating 10110101 left through carry (carry initially = 1). MSB (1) goes to carry flag, old carry (1) fills LSB position.

Right Shift (>>) with Carry

- Shifts all bits right by one position
- The rightmost bit (LSB) goes into the carry flag
- The previous carry flag value goes into the leftmost bit (LSB)
- Direction: LSB → Carry → MSB
- Useful for multi-precision arithmetic (working with numbers larger than register size), implementing rotate operations, serial data processing, checksum calculations



Example: Rotating 10110101 right through carry (carry initially = 0). LSB (1) goes to carry flag, old carry (0) fills MSB position.